

i Introduction

Department of Computer Science

Examination paper for: IT3708 Bio-Inspired Artificial Intelligence

Examination date: 2.6.2022

Examination time (from-to): 15:00 – 16:30

Permitted examination support material: A / All support material is allowed

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OTHER INFORMATION

Make your own assumptions: If a question is unclear/vague, make your own assumptions and specify them in your answer. Only contact academic contact in case of errors or insufficiencies in the question set.

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Citations: There is no requirement for citations, given the limited examination time. But if you think that it is natural and useful to include one or more citations on a question, feel free to do so.

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Weighting: Each "multiple choice" question is worth 1 point, for a total of 18 points. The "written answer" questions are worth a total of 12 points. For each "written answer" question, the maximum score is given with the question. The total number of points for the exam is 30.

ABOUT SUBMISSION

How to answer questions: All question types must be answered directly in Inspira. In Inspira, your answers are saved automatically every 15 seconds. **NB!** We advise against pasting content

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Withdrawing from the exam: If you become ill, or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Accessing your answer post-submission: You will find your answer in Archive when the examination time has expired.

1 crowding-02

Diversity preservation is important to promote exploration in the search space. Crowding is one way to do diversity preservation. Which of the following statements is **not** associated with the crowding approach?

Select one alternative:

- ☐ Crowding is useful for multimodal problems where we want to find multiple solutions in different parts of the search space
- ☒ A distance metric is not used as part of the crowding method ✔
- ☐ Group similar individuals and make them compete during survivor selection
- ☐ Parents and their offspring are grouped together in pairs, often consisting of one parent and one child, in order to keep the fittest among them with higher probability
- ☐ Set up a competition in which fitter individuals are selected for inclusion in the next generation with higher probability

Maximum marks: 1

2 fsharing-01

Diversity preservation is important to promote exploration in the search space. Fitness sharing is one way to do so. Which of the following statements is **not** associated with the fitness sharing approach?

Select one alternative:

- ☐ Individuals in the same niche compete for resources so their fitnesses are reduced compared to standard fitness computation
- ☐ Parents and their offspring are grouped together in pairs, each pair consisting of one parent and one child, in order to only keep the fittest among them 
- ☐ Individuals in more crowded niches are penalized more, or have their fitnesses reduced more, than those in less crowded niches
- ☐ Fitness sharing is useful for multimodal problems where we want to find multiple solutions in different parts of the search space
- ☐ Fitnesses are recalculated from raw fitnesses, for example in order to discourage the allocation of many individuals in niches near local optima

Maximum marks: 1

3 permutation-01

When working with individual representation for permutations, care is to be taken with the exploration operators like mutation and crossover. Which of the following statements is **not** a safe operation for permutation genotypes?

Select one alternative:

- ☐ Invert a subsequence of genes between two different positions of an individual
- ☐ Swap the alleles at two different loci of an individual
- ☐ Randomize a subsequence of genes between two different positions of an individual
- ☐ Use the partially mapped crossover (PMX) operator to copy over chromosome segments from parents to offspring
- ☐ Use a single crossover point to minimize conflicts in the generated offspring 

Maximum marks: 1

4 markov-01

Markov Chains (MC) are useful tools to analyze theoretical concepts of evolutionary algorithms. Consider a Simple Genetic Algorithm (SGA) in which the bit-string length is 8 and the population size is 6. Which of the following statements is **not** true?

Select one alternative:

- ☐ States that are similar can be grouped together to reduce the size of the MC model
- ☐ The population size and bit-string length impact directly the number of possible EA populations
- ☐ The search space cardinality is 64 ✓
- ☐ The total number of possible EA populations is 414,356,272,512
- ☐ The total number of possible EA populations makes MC analysis for this problem a challenge

Maximum marks: 1

5 MA-01

Memetic Algorithms (MA) are hybrid algorithms that integrate an Evolutionary Algorithm and some other algorithm, for example Stochastic Local Search. Which of the following statements is **not** true about MAs?

Select one alternative:

- ☐ To incorporate good solutions into the population, Stochastic Local Search can be used to climb from random individuals in the population
- ☐ When including known good solutions, average performance is often increased. However, best performance can come from a randomized inclusion of solutions
- ☐ Initializing populations with already known solutions is a common practice in MAs
- ☐ Constructive heuristics are commonly used, for example "Maximum number of conflicts first" in Constraint Optimization Problems
- ☐ Since Local Search considers neighborhoods, it can only be applied to permutation problems ✓

Maximum marks: 1

6 ACO-01

The Ant Colony Optimization (ACO) algorithm is an example of swarm intelligence that works well with routing problems, as it is robust and reliable. Which of the following is **not** true about ACO?

Select one alternative:

- ☐ In the first version of ACO, ants decide where to go using a probabilistic transition rule
- ☐ Changing explicit parameters (hyperparameters) for pheromone trail importance can have significant impact on ACO performance
- ☐ Variants of ACO for multi-objective optimization have not been developed yet 
- ☐ Pheromone updates increase the values associated with good solutions while also decreasing those that are associated with bad ones
- ☐ Edges that belong to short tours receive greater amount of pheromone and are therefore favored over long tours

Maximum marks: 1

7 continuous-01

Many applications in optimization are not discrete but continuous. Fortunately, Evolutionary Algorithms including Genetic Algorithms (GAs) can still be used. Which of the followings statements is **not** true when dealing with continuous optimization?

Select one alternative:

- ☐ Differential evolution and evolution strategies were designed for continuous optimization
- ☐ A mapping is always necessary to transform from real valued functions to binary genotypes and vice versa 
- ☐ Initialization as well as mutation typically sample from the domains using a probability distribution
- ☐ Genes typically take values from an interval on the real line
- ☐ Some of the *steps* of the continuous GA are similar to their binary counterparts, like fitness calculation, selection and crossover

Maximum marks: 1

8 DE-01

Differential Evolution (DE) is an interesting metaheuristic that works with real-valued vectors. Which of the following is **not** true about DE?

Select one alternative:

- ☐ In DE, crossover can be used
- ☐ Mutation occurs by creating a perturbation vector and adding it to a member of the population
- ☐ In DE/rand/1/bin, selection is deterministic and chooses between a pair of vectors: one from the original population and another from the trial vector population
- ☐ DE often uses three parameters (or hyperparameters), namely scaling factor, population size, and crossover probability
- ☐ DE was, when introduced, based on genetic algorithms 

Maximum marks: 1

9 selection-03

Parent selection is an important step in evolutionary algorithms as it helps to guide the direction of the search. Among these parent selection methods, which one **is** local?

Select one alternative:

- ☐ Linear rank-based selection
- ☐ Exponential rank-based selection
- ☐ Roulette-wheel selection
- ☐ Tournament selection 
- ☐ Fitness-proportionate selection

Maximum marks: 1

10 constraint-02

One way to use constraints in EAs is so-called indirect constraint handling. Which of the following statements is **not** a correct description of indirect constraint handling?

Select one alternative:

- ☐ Indirect constraint handling allows the EA to tune the fitness function by modifying weights while searching
- ☐ Indirect constraint handling can handle multiple constraints by including them in the penalty function
- ☐ Indirect constraint handling involves the use of a penalty function that is added to the original fitness function
- ☐ While both static and dynamic penalty functions are feasible, adaptive penalty functions have not been discussed in the literature ✓
- ☐ A distance metric is typically used in the penalty function of indirect constraint handling

Maximum marks: 1

11 constraint-03

There are different ways of handling constraints in EAs. In this question, we are concerned with their handling in a Constraint Satisfaction Problem (CSP), a Constrained Optimization Problem (COP), or a Free Optimisation Problem (FOP). Among the following statements, select the one that does **not** make sense when processing constraints with an EA.

Select one alternative:

- ☐ Adding a penalty function to the objective function, thus taking care of COP constraints, encourages the EA to ignore solutions that fall outside the feasible space
- ☐ Some constraints in a CSP can be captured by the EA's objective function, while the remaining constraints can be directly handled when solving the problem as a COP
- ☐ Constraints in a COP can be transformed into a CSP for which an EA can handle constraints directly ✓
- ☐ Constraints of a CSP can be handled indirectly by incorporating them into an FOP fitness function, which can be optimised using a traditional EA
- ☐ COP constraints can be tackled with indirect constraint handling, using a penalty function that is added to the original fitness function

Maximum marks: 1

12 phenotype-01

When discussing bio-inspired algorithms, one considers both phenotypic and genotypic concepts. Which of the following concepts deals with the phenotype as opposed to the genotype?

Select one alternative:

- ☐ Transmitted during reproduction
- ☐ Selection does not operate on it
- ☐ It is the genetic material of the individual
- ☐ Selection operates on it
- ☐ Affected by mutations



Maximum marks: 1

13 representation-02

The chromosome representation in an evolutionary algorithm is important, and typically a significant simplification from what is taking place in biology. Which of the following statements is **false** when dealing with genetic representation?

Select one alternative:

- ☐ The representation should be as general, abstract, and non-problem specific as possible
- ☐ To be most useful, the representation must capture a way to express an optimal or near-optimal solution to the problem
- ☐ The chromosome representation must be of appropriate size or dimension to be able to represent the problem parameters with enough granularity
- ☐ A representation must match genetic operators of recombination and mutation
- ☐ A good representation encodes relevant parameters of the problem

Maximum marks: 1

14 parameter-01

Setting parameters (or hyperparameters) is an important topic in bio-inspired AI. We often distinguish between parameter tuning and parameter control methods. Which of the following limitations or difficulties of parameter tuning is **incorrect**?

Select one alternative:

- ☐ Unfortunate parameter settings can be sources of errors or sub-optimal performance
- ☐ Even with “smart” tuning methods, parameter tuning often becomes expensive (timewise)
- ☐ Good values for previously set parameters could turn out to be bad during the real run
- ☐ Finding an optimal predetermined, time-varying schedule for changing the parameter is hard ✓
- ☐ Parameters are linked, so exhaustively searching for the optimal assignment to parameters is not feasible in general

Maximum marks: 1

15 NSGAI-02

NSGA-II is among the most prominent multi-objective EAs. Considering different parameters (or hyperparameters) and their effect on the behavior of multi-objective EAs, which of the following descriptions fits NSGA-II **best**?

Select one alternative:

- ☐ A sharing function parameter σ_{share} needs fixing in NSGA-II
- ☐ The NSGA-II algorithm requires fixing two parameters, namely: α_{share} and t_{dom}
- ☐ Two parameters, archive size (N) and depth (d), are important to tune for NSGA-II to work optimally
- ☐ NSGA-II seeks to achieve population diversity but not convergence
- ☐ Solutions compete with their crowding distances. No additional niching parameter is required to be set in NSGA-II ✓

Maximum marks: 1

16 PSO-01

Consider these bio-inspired algorithms: Particle Swarm Optimization (PSO) and Continuous Genetic Algorithm (CGA). Which of the following differences between PSO and CGA is **wrong**?

Select one alternative:

- ☐ PSO was introduced and is useful for continuous problems, while CGA is based on GAs designed for discrete settings
- ☐ Individuals in PSO (particles) do not share information directly, unlike individuals in CGAs where exchange of chromosome information plays a key role through crossover
- ☐ The metaphor is quite different: moving particles in PSO and evolution in CGA
- ☐ PSO did not originally use crossover while crossover typically is an important part of CGA
- ☐ PSO uses fitness for selection, unlike CGA which always uses a uniform probability distribution for selection ✓

Maximum marks: 1

17 swarm-01

Which of the following ideas is **not** considered as a key purpose or main principle in swarm intelligence?

Select one alternative:

- ☐ Individuals in the swarm transfer genetic information via crossover to the next gene ✓ on
- ☐ Even though some of the individuals get lost or make mistakes, the performance of the swarm is not dramatically affected
- ☐ Individuals in the swarm do not necessarily know the global state of the swarm
- ☐ The purpose of swarm intelligence is often to solve complex problems that a single individual could not solve alone
- ☐ Individuals in the swarm have local information, perform simple actions and have little to no memory

Maximum marks: 1

18 ES-02

An interesting approach for numerical optimization is the use of Evolution Strategies (ES). Which of the following statements is **not** true about ES?

Select one alternative:

- ☒ In the (μ, λ) -ES, the λ parents are replaced by the μ offspring ✓
- ☐ The idea of ES takeover time is to reflect how many generations it takes for the algorithm to fill the population with copies of the best individual
- ☐ Typically, parent selection in ES is unbiased in that every individual has the same probability of being selected
- ☐ In the $(1+1)$ -ES metaphor, chromosomes are vectors in $\mathbb{R}^n$ coupled with their strategy parameters
- ☐ The $(1+1)$ -ES is a metaphor that refers to one parent and one offspring every generation

Maximum marks: 1

19 convergence-A

On premature convergence: Why can premature convergence be a problem in Evolutionary Algorithms? Discuss **three** problems related to premature convergence (2 points max. per problem. 300 words max.)

Fill in your answer here

Format

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x_2

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Words: 0

Maximum marks: 6

20 parameter-B

The following is a description of an EA parameter (hyperparameter) setting method: "The scaling factor ϕ can be encoded in the chromosome, and the evolutionary algorithm can perform recombination and mutation on the chromosomes (including the ϕ -values) in the population. As a consequence, the ϕ -values leading to better individuals are more likely to survive."

Consider what type of EA parameter setting method is described in this quote. What are **three** limitations of this type of method? (2 points max. per limitation. 300 words max.)

Fill in your answer here

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Words: 0

Maximum marks: 6